

CSI247:
Data Structures

Lecture 08 – Packages

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Previous Lessons

- Sorting algorithm
  - Bubble sort
  - Insertion sort
  - Selection sort
  - Merge sort

Today …



Learning Objectives

- By the end of this, lecture you must be able to:
  - Understand what **packages** are
  - Understand various Java's built in packages
  - Understand and differentiate the 2 categories of packages
  - Create a user-defined package
  - Create a small Java program using a user-defined package

Introduction

Package: *A* group of related classes organized into a folder, which might have sub-folders

- Advantages
  - Promotes writing better and maintainable code
  - Prevent possibility of name conflicts
- Two categories:
  - Built-in Packages (refer to Java API)
    - Come bundled with the language
  - User-defined Packages
    - Must be created by programmers

a) Built-in packages

- Java's API is a **huge** library of pre-written classes
  - Rem: class is a blue-print of objects belonging to it
  - Examples of classes:
    - Scanner: input (e.g. keyboard)
    - PrintWriter: file output
    - StringTokenizer: breaking a String into tokens
  - Each of the above belongs to some package
  - The API is divided into packages



Java defined core packages

- Java has core packages
  - Each serves a different purpose
  - **You have used these packages!**
- Examples include:
  - java.lang: for Java's fundamental classes, e.g. String, Math
  - java.util: for collections framework and utility classes. e.g. Date, Random number generator, etc
  - java.io: for classes associated with file input and output
  - etc (See Java API documentation for more)

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How have we been using these packages?

- Add **import** keyword before your class definition
- Syntax:

  import packageName.className;
- E.g.
  - Import java.io.*;
- **Note**
  - **java.lang** package is imported by default in Java

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Example: using a built-in package

- Remember Scanner class for keyboard input?
  - It is in the **java.util** package

How have we been importing Scanner class

a) *import java.util.**; // import all classes in the package

b) *import java.util.**Scanner**; // import Scanner only

Which Scanner methods have we used?

- next(), nextInt(), nextLine(), hasNext(), etc

b) User-defined packages

- We can create our user defined packages
  - Create several user-defined classes
  - Bundle related classes together into a package
- Then import the package(s)/ relevant class(es) in our programs!

Steps?

a) Create the necessary directory structure
   - src directory: for java files
   - classes directory: class files and packages go here

b) Choose package name

- E.g. *pkg*
- Recommendation for package naming
  - Start package name with a lowercase letter
  - Use Caps at the beginning of each new word

c) Create source files (.java) under src folder

d) Insert keyword **package** followed by package name at the top of every source file to include in the package

- Syntax

```
package package_name;
```

illustration

```
package pkg;
public class A{
        // data & methods
}
```

- Save as A.java

```
package pkg;
public class B{
        // data & methods
}
```

- Save as B.java

Note:

- Both classes have statement **package pkg** at the top
  - They will be added to package **pkg**

d) Create the package

How?

- Compile each class to be included in the package, individually

javac –d classes/ -sourcepath src src/…/class_name;

Note:

- Compile from parent directory of src and classes folder
- -d: create directory with package_name indicated in source files
- classes: package_name destination (or .(dot) for current directory)

- Upon successful compilation:

- **classes** directory will have a sub-directory called **pkg**
  - **pkg** will have a class file for each compiled .java file

e) Using the package in a program:

- Insert **import** keyword (as 1st line), followed by *package_name.class_name* in Tester class

- Syntax

```
import package_name.class_name;
```

- Example

```
import pkg.A;  // or import pkg.*; (for all classes)
public class pkgTester{
 // …
}
```

e) Testing if the package works properly

How?

- Executing the Tester class
  - Use this command:

java –classpath classes packagefolder.Tester_name;

Note:

- You must work from folder above classes folder



Summary

- Packages
  - Built-in packages
  - User-defined packages
    - Declaration
    - Using them



Next lesson

- OOP

Q & A

